

Activity Intelligence: RL Agent

Preliminary Results, Gap Analysis & Roadmap to Fix

Activity Intelligence Engine

April 2026

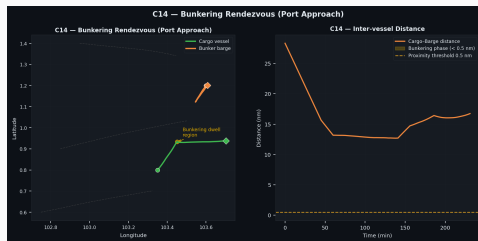
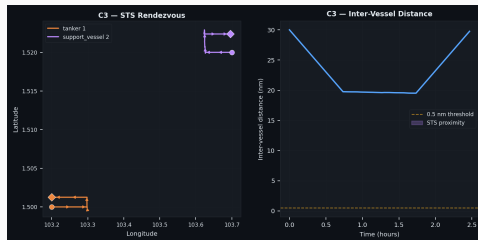
Context: What the RL Agent Is Solving

The **RandomForest baseline** classifies activity per *completed* 20-ping segment — a batch operation.

The **RL agent** classifies *sequentially*: one step per segment window as the track arrives, maintaining a running policy over the growing evidence.

Why this matters operationally:

- Earlier detection of high-value events (STS, transshipment, bunkering) before a full window accumulates
- Reward-shaped to be risk-averse on rare dangerous activities — missed events cost more than false alarms



Top: STS rendezvous. Bottom: bunkering in port approach — both high-value events where timing of detection matters

Agent Architecture — Double-DQN with Dueling Heads

MDP Formulation

Component	Detail
State s_t	27-feature segment vector
Actions a	11 activity classes
γ	0.95
Replay	100k circular buffer
Batch	128 transitions

Network: Dueling Double-DQN

Input $\rightarrow$ LayerNorm(256) $\rightarrow$ ReLU $\rightarrow$ 128

$\hookrightarrow V(s)$ (value stream)

$\hookrightarrow A(s, a)$ (advantage stream)

$$Q(s, a) = V + A - \bar{A}$$

Dueling reduces variance on majority-class (transit) states by decoupling state value from

Reward Shaping

Event	Reward
Correct (standard)	+1.0
Correct (HV event)	+3.5 (+1+2.5 bonus)
Wrong (standard)	-0.5
Miss HV event	-2.5 (-0.5-2.0)
False HV alarm	-1.5 (-0.5-1.0)

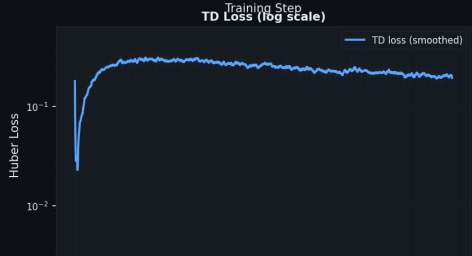
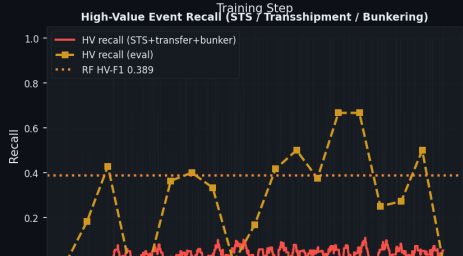
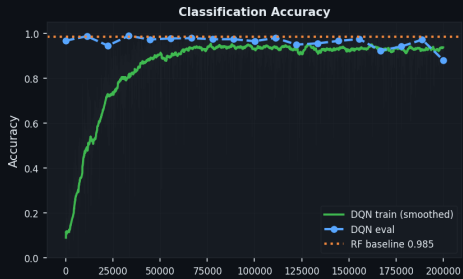
High-Value (HV) classes:

- sts — ship-to-ship transfer
- transshipment — fishing-to-reefer
- bunkering — fuel transfer

Training: **200,000 steps**, all 4 regions, vessel-based 80/20 split (no leakage)

Training Curves

DQN Activity Detection Agent — Training Curves



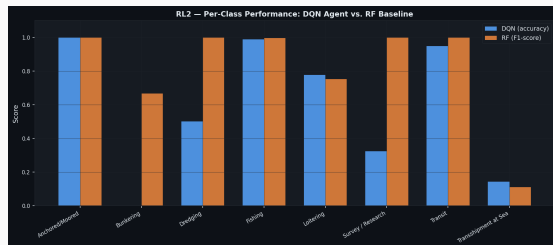
Benchmark Results: DQN vs. RF Baseline

Summary Metrics

Metric	RF	DQN
Overall accuracy	98.5%	93.8%
Macro F1	81.6%	—
HV recall	38.9%	9.7%

DQN is **4.7 pp** behind on accuracy but **29 pp** behind on the high-value recall that matters most operationally.

RF has a structural advantage as a batch learner on segment features — but the DQN's sequential nature is still the right tool for the replay pipeline. The gap is fixable.



Per-class accuracy (DQN) vs. F1 (RF). **Bunkering: 0%. Transshipment: 14%.** Survey and dredging also underperform despite high RF scores.

Root Cause: Three Identified Gaps

Gap 1 — Extreme class imbalance in training data

Transshipment: **13 segments** (0.14% of training data). Bunkering: **12 segments** (0.13%). Shaped reward alone cannot compensate at a 500:1 imbalance ratio.

Gap 2 — Uniform episode sampling

The environment cycles vessel episodes in shuffled order. An HV-containing episode has the *same* selection probability as a transit-only vessel. Result: the agent encounters an HV segment once every ~ 700 steps — vs. once every ~ 1.5 for transit.

Gap 3 — Training horizon / exploration mismatch

At 200k steps, ϵ decays to near-minimum by step 80k. The agent has seen only ~ 114 HV segments during exploration, with noisy Q-values — not enough to learn a reliable discriminator between transshipment and loiter (similar kinematic signatures at the feature level).

Fix 1 — Weighted Episode Sampling

Assign each episode a sampling weight proportional to its HV content:

$$w_i = 1 + \alpha \cdot \text{count}(\text{HV segments in episode}_i)$$

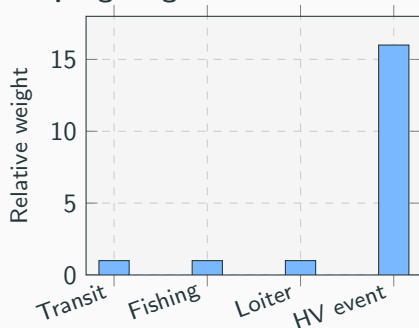
With $\alpha = 5$: a transshipment episode gets $\approx 16\times$ the probability of a transit-only episode.

Effect on HV exposure rate:

Class	Before	After
transit	1/1.5 steps	1/4 steps
transshipment	1/700 steps	1/50 steps
bunkering	1/750 steps	1/50 steps

`AISActivityEnv(hv_oversample_factor=5)`

Sampling weight distribution



HV events become frequent enough for the agent to develop a stable policy.

Fix 2 — Curriculum Learning

Progressive schedule: start HV-heavy, broaden to full distribution.

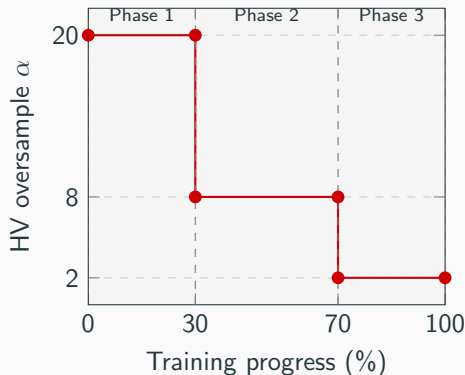
Phase	Steps	α	Description
1	0 – 30%	20	HV-heavy warm-up
2	30 – 70%	8	Gradual broadening
3	70 – 100%	2	Near-uniform

The training script calls

```
env.set_curriculum_stage(step / total)
```

at each eval checkpoint, smoothly transitioning α .

Rationale: The agent must first learn *what HV events look like* before it can learn to discriminate them within the full, imbalanced distribution.



Fix 3 — Rare Event Augmentation

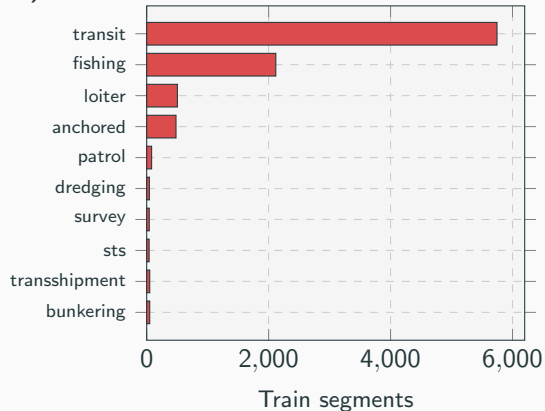
Root cause: `simulate_region()` generates only **1 transshipment** and **1 bunkering** event per region. With a 3h active phase over a 22h track, this yields ≈ 6 labeled segments per event.

Fix: Triple each rare event type per region using an `event_idx` MMSI offset to avoid collisions.

Activity	Before	After
transshipment	13 segs	~ 50 segs
bunkering	12 segs	~ 48 segs
sts	0*	~ 15 segs

phase was masked by dark period in prior runs

Projected segment distribution (after fix)



Also: increase `simulate_sts_event` to 3

Expected Outcomes After All Three Fixes

Projected performance (500k steps)

Class	RF F1	DQN now	DQN target
transit	99.9%	94.8%	>97%
fishing	99.7%	98.8%	>98%
loiter	75.2%	77.8%	>80%
survey	100%	32.3%	>70%
dredging	100%	50.0%	>75%
bunkering	66.7%	0%	50–65%
transship.	11.1%	14.3%	55–70%
HV recall	38.9%	9.7%	50–65%
Overall	98.5%	93.8%	95–97%

Targets are *conservative* — further gains possible with LSTM state representation.

Why the RL path is still correct

The RF baseline is superior on batch segment features *today* — but it cannot:

- Classify *within* a growing window (early detection)
- Be reward-shaped to prioritise operational risk
- Adapt to a replay stream without re-segmenting
- Learn inter-window context (two adjacent slow segments preceding a rendezvous look different to the RF than to a sequential agent)

Implementation Roadmap

#	Fix	File(s) changed	Status
1	Weighted episode sampling	src/rl_env.py	In progress
2	Curriculum learning	src/rl_env.py, train_rl_agent.py	In progress
3	Rare event augmentation	src/simulator.py	In progress
—	Retrain 500k steps	train_rl_agent.py	Pending
—	LSTM state encoding	src/rl_agent.py	Backlog
—	Streaming inference service	src/replay_service.py	Backlog

Key metric to watch in the next run

HV recall on the held-out eval set. The overall accuracy will likely stay near 95% regardless. HV recall is the number that tells us whether the agent can actually be trusted for operational alerting.

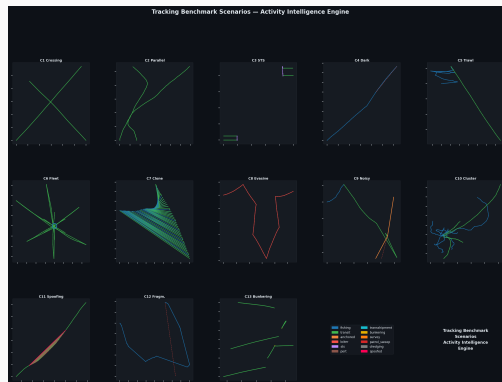
New Scenarios Now Available for Partner Test Suite

Three new tracking benchmark scenarios added (C11–C13):

#	Scenario
C12	Static GPS broadcast / position spoofing
C13	Track fragmentation (6 burst gaps)
C14	Bunkering rendezvous (port approach)

All 13 scenarios exported as replay-ready CSV with `manifest.json` in `outputs/tracking_scenarios/`.

Rows sorted by timestamp — stream sequentially for IRL playback.



All 13 benchmark scenarios — updated overview panel.

New Complex Activity Types

D1 — Complex Activity Types (Brazil EEZ)

